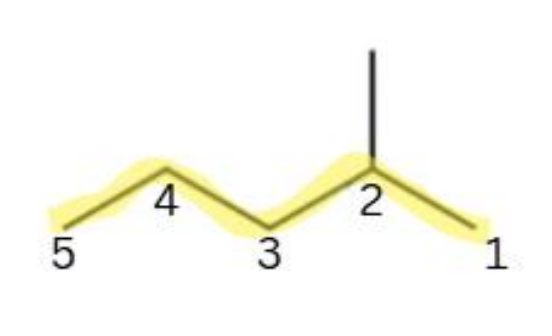
PART A · STRUCTURE → NAME · FOUNDATION

CARD A01

Molecular formula: C_6H_{14}

UPPER LEFT

1. Numbered parent



UPPER RIGHT

2. Naming trail

Parent: the longest chain is 5 C, so pentane.
Numbering: a methyl sits on the second carbon from one end (2), but on the fourth from the other (4). 2 is lower.
Substituents: methyl on C2.

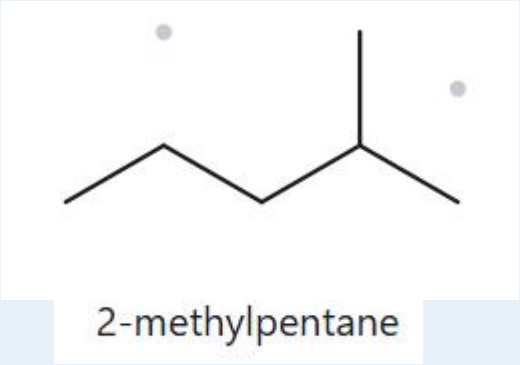
LOWER LEFT

3. NAME

2-methylpentane

LOWER RIGHT

4. The Alkane check



MATCH

VERIFICATION STRIP

☐ longest chain/ring ☐ lowest locants ☐ alphabetical order ☐ punctuation ☐ formula counted

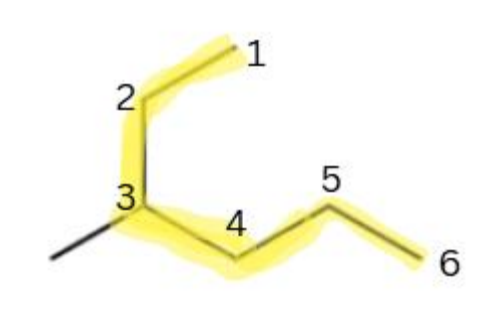
PART A · STRUCTURE → NAME · FOUNDATION

CARD A02

Molecular formula: C_7H_{16}

UPPER LEFT

1. Numbered parent



UPPER RIGHT

2. Naming trail

Parent: the longest chain bends through the branch point, going from the end of the ethyl arm to the end of the 3-carbon arm. That is 6 C, so hexane.
Numbering: from the ethyl end, the branch carbon is C3. From the other end, it is C4. 3 is lower.
Substituents: methyl on C3. (The one-carbon arm is the methyl.)

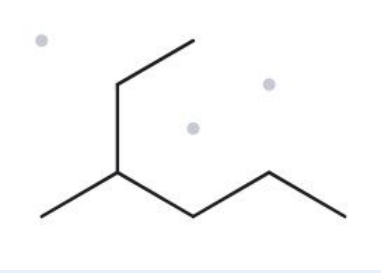
LOWER LEFT

3. NAME

3-methylhexane

LOWER RIGHT

4. The Alkane check



3-methylhexane



MATCH

VERIFICATION STRIP

☐ longest chain/ring ☐ lowest locants ☐ alphabetical order ☐ punctuation ☐ formula counted

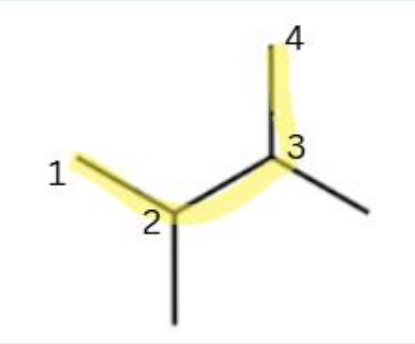
PART A · STRUCTURE → NAME · FOUNDATION

CARD A03

Molecular formula: C_6H_{14}

UPPER LEFT

1. Numbered parent



UPPER RIGHT

2. Naming trail

Parent: the longest chain is 4 C, so butane. The two adjacent branch carbons are C2 and C3.
Numbering: {2,3} from either end.
Substituents: methyl on C2 and methyl on C3, so use di-.

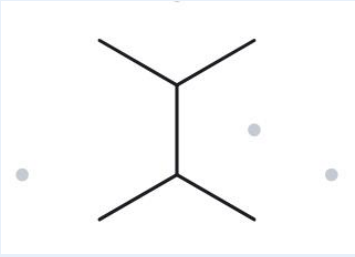
LOWER LEFT

3. NAME

2,3-dimethylbutane

LOWER RIGHT

4. The Alkane check



2,3-dimethylbutane

MATCH

VERIFICATION STRIP

☐ longest chain/ring ☐ lowest locants ☐ alphabetical order ☐ punctuation ☐ formula counted

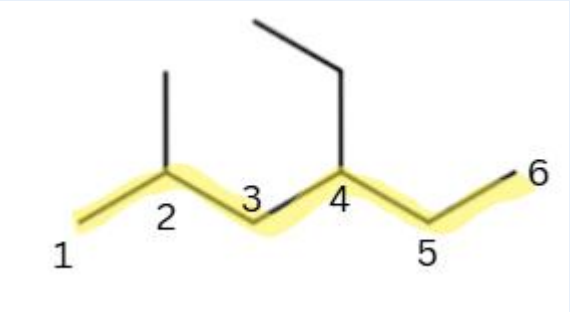
PART A · STRUCTURE → NAME · DEVELOPING

CARD A04

Molecular formula: C_9H_{20}

UPPER LEFT

1. Numbered parent



UPPER RIGHT

2. Naming trail

Parent: the longest chain is 6 C (several equal paths exist), so hexane. All of them carry one methyl and one ethyl.
Numbering: one direction gives {2,4}, the other gives {3,5}. 2 < 3, so use {2,4}.
Substituents: methyl on C2, ethyl on C4. Ethyl is cited first alphabetically.

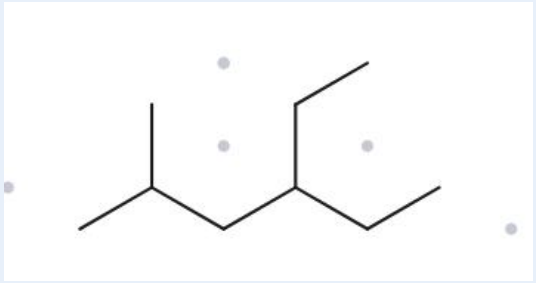
LOWER LEFT

3. My name (written first)

4-ethyl-2-methylhexane

LOWER RIGHT

4. The Alkane check



4-ethyl-2-methylhexane

MATCH

VERIFICATION STRIP

☐ longest chain/ring ☐ lowest locants ☐ alphabetical order ☐ punctuation ☐ formula counted

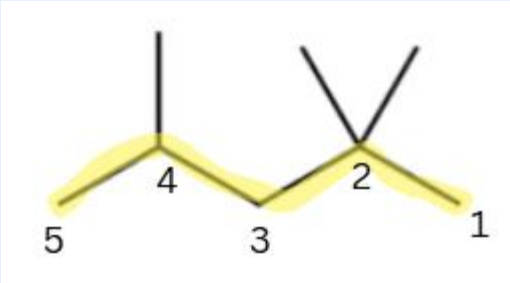
PART A · STRUCTURE → NAME · DEVELOPING

CARD A05

Molecular formula: C_8H_{18}

UPPER LEFT

1. Numbered parent



UPPER RIGHT

2. Naming trail

Parent: the longest chain is 5 C, so pentane.
Numbering: one direction gives {2,2,4}, the other gives {2,4,4}. At the second term, 2 < 4.
Substituents: two methyls on C2 and one methyl on C4, so use tri-.

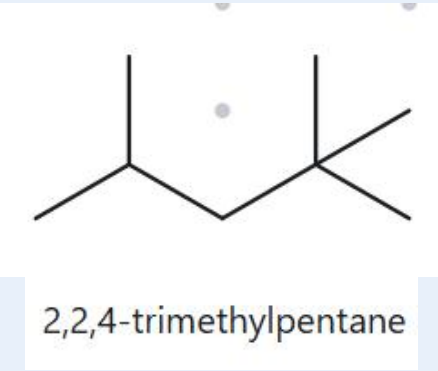
LOWER LEFT

3. NAME

2,2,4-trimethylpentane

LOWER RIGHT

4. The Alkane check



MATCH

VERIFICATION STRIP

☐ longest chain/ring ☐ lowest locants ☐ alphabetical order ☐ punctuation ☐ formula counted

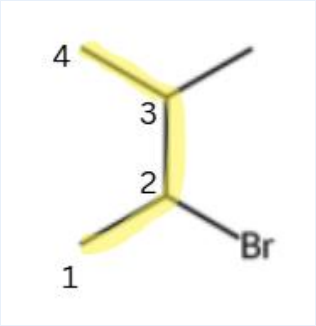
PART A · STRUCTURE → NAME · DEVELOPING

CARD A06

Molecular formula: $C_5H_{11}Br$

UPPER LEFT

1. Numbered parent



UPPER RIGHT

2. Naming trail

Parent: the longest chain is 4 C, so butane.
Numbering: {2,3} from either end, so it is a tie. Bromo comes before methyl alphabetically, so bromo gets C2.
Substituents: bromo on C2, methyl on C3.

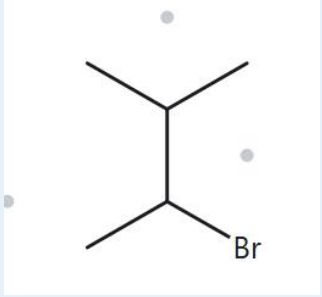
LOWER LEFT

3. NAME

2-bromo-3-methylbutane

LOWER RIGHT

4. The Alkane check



2-bromo-3-methylbutane

MATCH

VERIFICATION STRIP

☐ longest chain/ring ☐ lowest locants ☐ alphabetical order ☐ punctuation ☐ formula counted

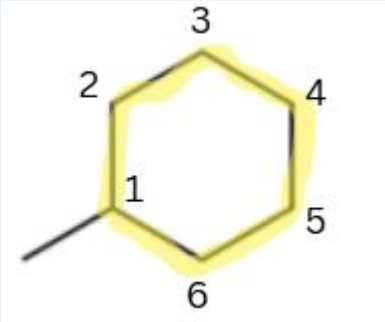
PART A · STRUCTURE → NAME · DEVELOPING

CARD A07

Molecular formula: C_7H_{14}

UPPER LEFT

1. Numbered parent



UPPER RIGHT

2. Naming trail

Parent: the ring is the parent, with 6 C, so cyclohexane.
Numbering: only one substituent, so no locant is needed.
Substituents: methyl.

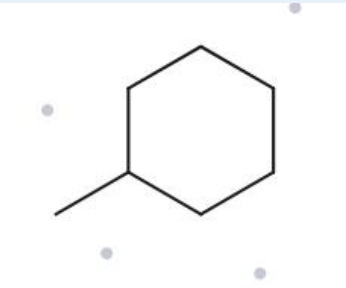
LOWER LEFT

3. NAME

methylcyclohexane

LOWER RIGHT

4. The Alkane check



methylcyclohexane



VERIFICATION STRIP

☐ longest chain/ring ☐ lowest locants ☐ alphabetical order ☐ punctuation ☐ formula counted

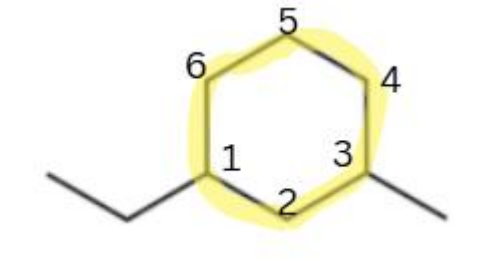
PART A · STRUCTURE → NAME · CHALLENGE

CARD A08

Molecular formula: C_9H_{18}

UPPER LEFT

1. Numbered parent



UPPER RIGHT

2. Naming trail

Parent: the ring, cyclohexane.
Numbering: the two substituents are separated by one ring carbon, so the locants are {1,3} either way round. Tie, so alphabetical order decides. Ethyl gets C1.
Substituents: ethyl on C1, methyl on C3.

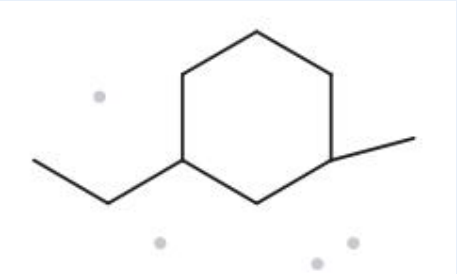
LOWER LEFT

3. NAME

1-ethyl-3-methylcyclohexane

LOWER RIGHT

4. The Alkane check



1-ethyl-3-methylcyclohexane

MATCH

VERIFICATION STRIP

☐ longest chain/ring ☐ lowest locants ☐ alphabetical order ☐ punctuation ☐ formula counted

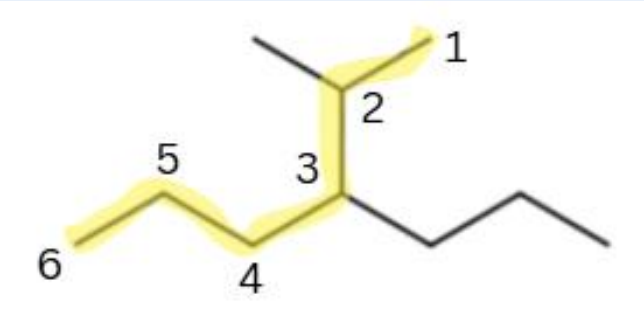
PART A · STRUCTURE → NAME · CHALLENGE

CARD A09

Molecular formula: $C_{10}H_{22}$

UPPER LEFT

1. Numbered parent



UPPER RIGHT

2. Naming trail

Parent: trace from one methyl of the isopropyl group, through its CH and the branch carbon, then down the right-hand side. That is 6 C, so hexane.
Substituents: the other methyl of the isopropyl group is on the second carbon of this chain. The left-hand arm of the original zigzag (3 carbons) is a propyl on the third carbon.
Numbering: from the isopropyl end the locants are {2,3}. From the other end they are {4,5}. {2,3} wins.
Alphabetical order: methyl before propyl.

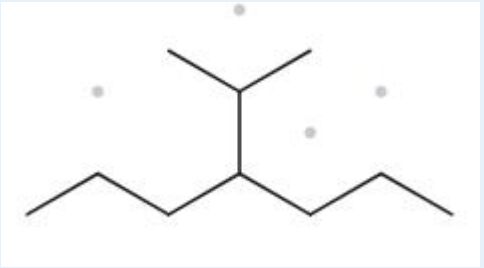
LOWER LEFT

3. NAME

2-methyl-3-propylhexane

LOWER RIGHT

4. The Alkane check



2-methyl-3-propylhexane

MATCH

VERIFICATION STRIP

☐ longest chain/ring ☐ lowest locants ☐ alphabetical order ☐ punctuation ☐ formula counted

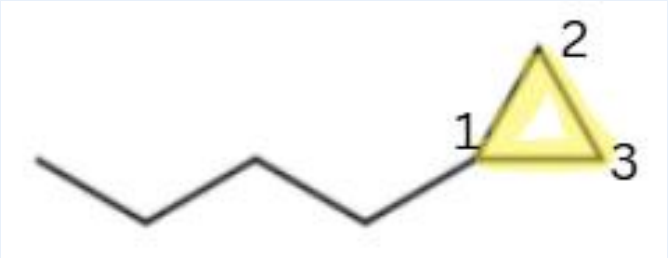
PART A · STRUCTURE → NAME · CHALLENGE

CARD A10

Molecular formula: C_8H_{16}

UPPER LEFT

1. Numbered parent



UPPER RIGHT

2. Naming trail

Parent: the ring is the parent, with 3 C, so cyclopropane. The straight tail has 5 C, which is longer than the ring, but this activity always takes the ring as the parent.
Numbering: only one substituent, so no locant is needed.
Substituents: a 5-carbon straight chain, pentyl.

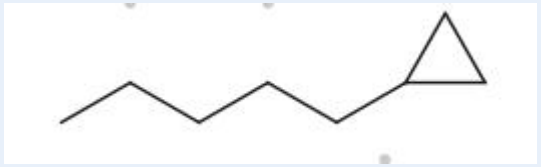
LOWER LEFT

3. NAME

pentylcyclopropane

LOWER RIGHT

4. The Alkane check



pentylcyclopropane

MATCH

VERIFICATION STRIP

☐ longest chain/ring ☐ lowest locants ☐ alphabetical order ☐ punctuation ☐ formula counted

PART B · NAME → STRUCTURE · FOUNDATION

CARD B01

Given name: 2-methylpropane

UPPER LEFT

1. Name decoded

Parent: propane → a 3-carbon chain.
Substituents: methyl on C2.
Expected formula: 3 + 1 = 4 C; open chain → C_4H_{10}

UPPER RIGHT

2. Bond-line structure (The Alkane)



LOWER LEFT

3. Formula check

Counted from the drawing: C_4H_{10}
Open chain: $C_n \quad H_{2n} \quad _{+2} = C_4H_{10}$ ✓

LOWER RIGHT

4. Back-check

Returned by The Alkane: 2-methylpropane. ✓ Same as the given name.
✓ parent length matches | ✓ substituent positions match | ✓ formula agrees | ✓ back-check recorded

2-methylpropane

VERIFICATION STRIP

☐ **parent length** ☐ **substituent positions** ☐ **formula agrees** ☐ **back-check recorded**

PART B · NAME → STRUCTURE · FOUNDATION

CARD B02

Given name: 3-ethylpentane

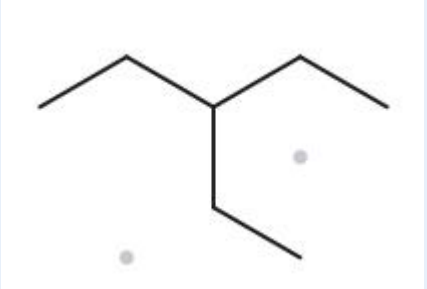
UPPER LEFT

1. Name decoded

Parent: pentane → a 5-carbon chain.
Substituents: ethyl on C3.
Expected formula: 5 + 2 = 7 C; open chain → C_7H_{16}

UPPER RIGHT

2. Bond-line structure (The Alkane)



LOWER LEFT

3. Formula check

Counted from the drawing: C_7H_{16}
Open chain: $C_n \quad H_{2n} \quad _{+2} = C_7H_{16}$ ✓

LOWER RIGHT

4. Back-check

Returned by The Alkane: 3-ethylpentane. ✓ Same as the given name.
parent length matches | ✓ substituent positions match | ✓ formula agrees | ✓ back-check recorded

3-ethylpentane

VERIFICATION STRIP

☐ **parent length** ☐ **substituent positions** ☐ **formula agrees** ☐ **back-check recorded**

PART B · NAME → STRUCTURE · FOUNDATION

CARD B03

Given name: 2,2-dimethylbutane

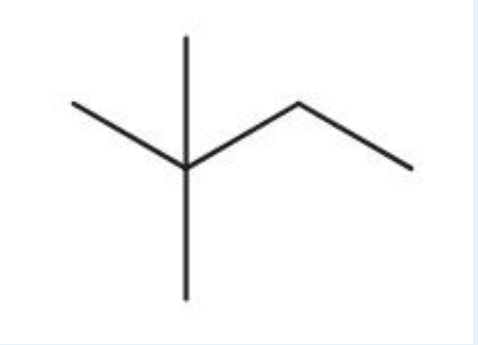
UPPER LEFT

1. Name decoded

Parent: butane → a 4-carbon chain.
Substituents: two methyls (dimethyl) on C2.
Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}

UPPER RIGHT

2. Bond-line structure (The Alkane)



LOWER LEFT

3. Formula check

Counted from the drawing: C_6H_{14}
Open chain: $C_n \quad H_{2n} \quad _{+2} = C_6H_{14}$ ✓

LOWER RIGHT

4. Back-check

Returned by The Alkane: 2,2-dimethylbutane. ✓ Same as the given name.
✓ parent length matches | ✓ substituent positions match | ✓ formula agrees | ✓ back-check recorded

2,2-dimethylbutane

VERIFICATION STRIP

☐ **parent length** ☐ **substituent positions** ☐ **formula agrees** ☐ **back-check recorded**

PART B · NAME → STRUCTURE · DEVELOPING

CARD B04

Given name: 3-ethyl-2-methylpentane

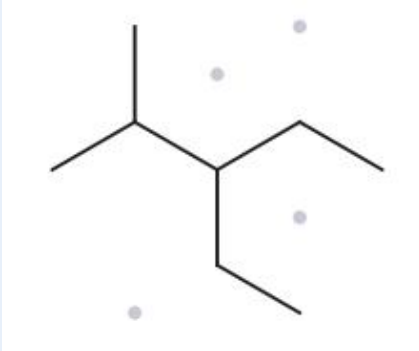
UPPER LEFT

1. Name decoded

Parent: pentane → a 5-carbon chain.
Substituents: methyl on C2; ethyl on C3.
Expected formula: 5 + 1 + 2 = 8 C; open chain → C_8H_{18}

UPPER RIGHT

2. Bond-line structure (The Alkane)



LOWER LEFT

3. Formula check

Counted from the drawing: C_8H_{18}
Open chain: $C_n \quad H_{2n} \quad _{+2} = C_8H_{18}$ ✓

LOWER RIGHT

4. Back-check

Returned by The Alkane: 3-ethyl-2-methylpentane. ✓
Same as the given name.
✓ parent length matches | ✓ substituent positions match | ✓ formula agrees | ✓ back-check recorded

3-ethyl-2-methylpentane

VERIFICATION STRIP

☐ **parent length** ☐ **substituent positions** ☐ **formula agrees** ☐ **back-check recorded**

PART B · NAME → STRUCTURE · DEVELOPING

CARD B05

Given name: 1-chloro-2-methylpropane

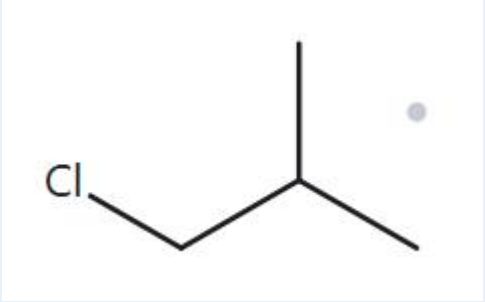
UPPER LEFT

1. Name decoded

Parent: propane → a 3-carbon chain.
Substituents: chloro on C1; methyl on C2.
Expected formula: 3 + 1 = 4 C; open chain → C_4H_{10}; Cl replaces one H → C_4H_9Cl

UPPER RIGHT

2. Bond-line structure (The Alkane)



LOWER LEFT

3. Formula check

Counted from the drawing: C_4H_9Cl
Open chain: C_nH_{2n+2} = C_4H_{10}; minus one H for Cl = C_4H_9Cl ✓

LOWER RIGHT

4. Back-check

Returned by The Alkane; 1-chloro-2-methylpropane. ✓
Same as the given name.
✓ parent length matches | ✓ substituent positions match | ✓ formula agrees | ✓ back-check recorded

1-chloro-2-methylpropane

VERIFICATION STRIP

☐ **parent length** ☐ **substituent positions** ☐ **formula agrees** ☐ **back-check recorded**

PART B · NAME → STRUCTURE · DEVELOPING

CARD B06

Given name: 1,2-dimethylcyclopentane

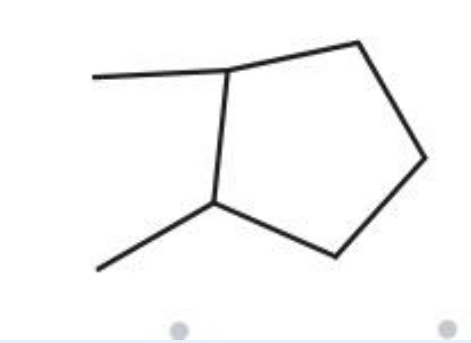
UPPER LEFT

1. Name decoded

Parent: cyclopentane → a 5-carbon ring.
Substituents: methyl on C1; methyl on C2.
Expected formula: 5 + 2 = 7 C; one ring → C_7H_{14}

UPPER RIGHT

2. Bond-line structure (The Alkane)



LOWER LEFT

3. Formula check

Counted from the drawing: C_7H_{14}
One ring: C_n H_{2n} = C_7H_{14} ✓

LOWER RIGHT

4. Back-check

Returned by The Alkane: 1,2-dimethylcyclopentane. ✓
Same as the given name.
✓ parent length matches | ✓ substituent positions match | ✓ formula agrees | ✓ back-check recorded

1,2-dimethylcyclopentane

VERIFICATION STRIP

☐ **parent length** ☐ **substituent positions** ☐ **formula agrees** ☐ **back-check recorded**

PART B · NAME → STRUCTURE · DEVELOPING

CARD B07

Given name: 4-ethyl-3,3-dimethylheptane

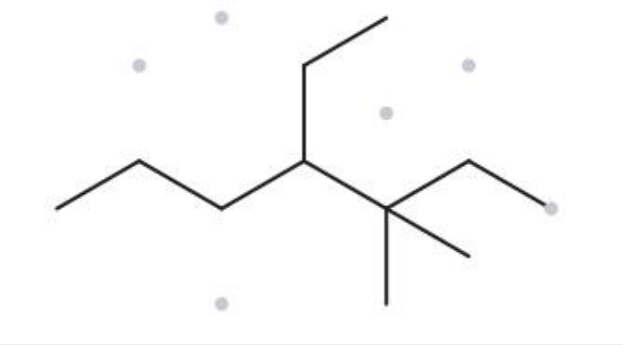
UPPER LEFT

1. Name decoded

Parent: heptane → a 7-carbon chain.
Substituents: two methyls (dimethyl) on C3; ethyl on C4.
Expected formula: 7 + 2 + 2 = 11 C; open chain → $C_{11}H_{24}$

UPPER RIGHT

2. Bond-line structure (The Alkane)



LOWER LEFT

3. Formula check

Counted from the drawing: $C_{11}H_{24}$
Open chain: $C_n \quad H_{2n} \quad _{+2} = C_{11}H_{24}$ ✓

LOWER RIGHT

4. Back-check

4-ethyl-3,3-dimethylheptane. ✓ Same as the given name.
✓ parent length matches | ✓ substituent positions match | ✓ formula agrees | ✓ back-check recorded

4-ethyl-3,3-dimethylheptane

VERIFICATION STRIP

☐ **parent length** ☐ **substituent positions** ☐ **formula agrees** ☐ **back-check recorded**

PART B · NAME → STRUCTURE · CHALLENGE

CARD B08

Given name: tert-butylcyclohexane

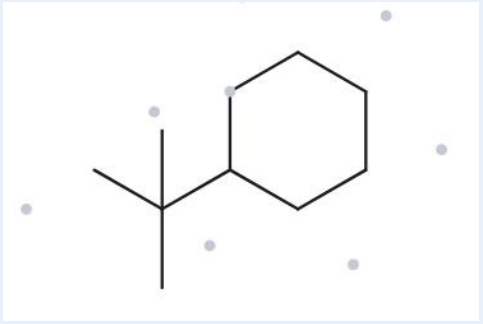
UPPER LEFT

1. Name decoded

Parent: cyclohexane → a 6-carbon ring.
Substituents: tert-butyl on C1 (one carbon bearing three methyls = 4 C).
Expected formula: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$

UPPER RIGHT

2. Bond-line structure (The Alkane)



LOWER LEFT

3. Formula check

Counted from the drawing: $C_{10}H_{20}$
One ring: $C_n \quad H_{2n} \quad = C_{10}H_{20}$ ✓

LOWER RIGHT

4. Back-check

The Alkane will likely return (1,1-dimethylethyl)cyclohexane. ✓ Same compound as the given name, written with a different accepted name for the group (Appendix B.3). Say this on the card.
✓ parent length matches | ✓ substituent positions match | ✓ formula agrees | ✓ back-check recorded

(1,1-dimethylethyl)cyclohexane

VERIFICATION STRIP

☐ **parent length** ☐ **substituent positions** ☐ **formula agrees** ☐ **back-check recorded**

PART B · NAME → STRUCTURE · CHALLENGE

CARD B09

Given name: 3-ethyl-1,1-dimethylcyclohexane

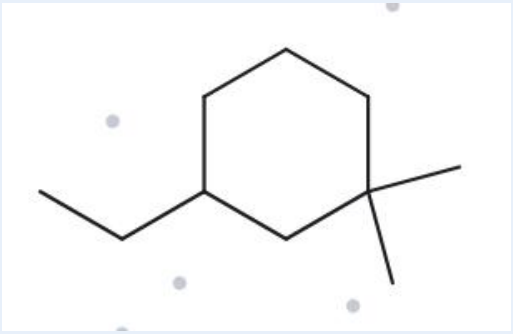
UPPER LEFT

1. Name decoded

Parent: cyclohexane → a 6-carbon ring.
Substituents: two methyls (dimethyl) on C1; ethyl on C3.
Expected formula: 6 + 2 + 2 = 10 C; one ring → $C_{10}H_{20}$

UPPER RIGHT

2. Bond-line structure (The Alkane)



LOWER LEFT

3. Formula check

Counted from the drawing: $C_{10}H_{20}$
One ring: $C_n \quad H_{2n} \quad = C_{10}H_{20}$ ✓

LOWER RIGHT

4. Back-check

3-ethyl-1,1-dimethylcyclohexane. ✓ Same as the given name. The locant set {1,1,3} beats {1,3,3}.
✓ parent length matches | ✓ substituent positions match | ✓ formula agrees | ✓ back-check recorded

3-ethyl-1,1-dimethylcyclohexane

VERIFICATION STRIP

☐ **parent length** ☐ **substituent positions** ☐ **formula agrees** ☐ **back-check recorded**

PART B · NAME → STRUCTURE · CHALLENGE

CARD B10

Given name: 2-ethylbutane (deliberately incorrect name)

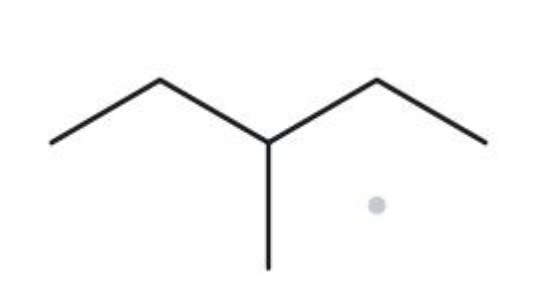
UPPER LEFT

1. Name decoded

Parent: butane → a 4-carbon chain.
Substituents: ethyl on C2.
Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}

UPPER RIGHT

2. Bond-line structure (The Alkane)



LOWER LEFT

3. Formula check

Counted from the drawing: C_6H_{14}
Open chain: $C_n \quad H_{2n} \quad _{+2} = C_6H_{14}$ ✓

LOWER RIGHT

4. Back-check

3-methylpentane, so this is ✗ against the given name.
Correct name: 3-methylpentane.
Rule broken: the parent must be the longest continuous chain. The ethyl branch extends the chain to 5 carbons, and the leftover carbon becomes a methyl on C3.
✓ parent length matches | ✓ substituent positions match | ✓ formula agrees | ✓ back-check recorded (record both names)

3-methylpentane

VERIFICATION STRIP

☐ **parent length** ☐ **substituent positions** ☐ **formula agrees** ☐ **back-check recorded**